A WEAK TOPOLOGY ON METRIC SPACES

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Abstract. We use metric functionals to construct a topology on metric spaces that induces d-weak convergence. We show that this topology is invariant under surjective isometries and characterize its convergence in terms of weak metrics. We prove that the topology is T_1 on every metric space. For normed linear spaces, we further show that it is Hausdorff and coarser than the classical weak topology. Finally, we construct a sequence in $C[0,1]$ that is unbounded in the sup-norm yet converges d-weakly, thereby correcting an assertion in an earlier work.

1. Main Results

In an earlier work [2], we introduced the notion of d-weak convergence in arbitrary metric spaces (X,d). We restricted our analysis there to sequences with the purpose of testing and validating our ideas in a simple way. We verified that this notion behaves as expected in *all* normed linear spaces; every *bounded sequence* in a normed linear space converges d-weakly if and only if it converges in the classical weak sense.

Our goal here is to investigate the topological structure underlying this notion of convergence. We begin by recalling its formal definition.

Definition 1.1. Let (X,d) be a metric space, and let $X^\diamondsuit$ be the space of all metric functionals on X. We say that a net of points $(x_i)_{i\in I}$ in X converges d-weakly to a point $x \in X$ if for every $\mathbf{h} \in X^\diamondsuit$ we have

$$\liminf_{i\in I} \mathbf{h}(x_i) \geq \mathbf{h}(x).$$

Our first result shows that d-weak convergence is induced by a topology.

Theorem 1.2. *Let (X,d) be a metric space and $X^\diamondsuit$ the space of all metric functionals on X. Then, there exists a topology on X induced by $X^\diamondsuit$, denoted by $\sigma(X,X^\diamondsuit)$, with the following properties:*

1. *The topology $\sigma(X,X^\diamondsuit)$ is coarser than the metric topology on X.*
2. *The topology $\sigma(X,X^\diamondsuit)$ on X makes every metric functional lower semi-continuous.*
3. *A net in X converges in the topology $\sigma(X,X^\diamondsuit)$ if and only if it converges d-weakly.*

In [2, Conjecture 1.5], we conjectured that d-weakly convergent sequences in normed linear spaces must be bounded. The conjecture was motivated by the results stated in [2, Theorem 1.4]. However, the assertion concerning $C[0,1]$ in that theorem is incorrect. The following theorem shows that the conjecture is false.

Theorem 1.3. *There exists a sequence (f_n) in $C[0,1]$ that converges to 0 in the topology $\sigma(C[0,1],C[0,1]^\diamondsuit)$ and $\|f_n\|_\infty \to \infty$.*

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We next establish an invariance property of the topology $\sigma(X, X^\diamondsuit)$ under surjective isometries.

Theorem 1.4. *Let (X, d) and (Y, D) be metric spaces. Suppose that $T : X \to Y$ is a distance-preserving mapping, that is, $D(Tp, Tq) = d(p, q)$ for all $p, q \in X$. Assume that $Y = TX$. Then, the mapping*

$$T : (X, \sigma(X, X^\diamondsuit)) \to (Y, \sigma(Y, Y^\diamondsuit))$$

is a homeomorphism.

The terminology *weak topology on metric spaces* is further justified by a connection with the notion of weak metrics, introduced by H. Ribeiro [5] in 1943. We begin by recalling the relevant definition.

Definition 1.5. Let X be a nonempty set. A mapping $\delta : X \times X \to [0, +\infty)$ is called a *weak metric* if for every $x, y, z \in X$ we have

$$\delta(x, x) = 0 \quad \text{and} \quad \delta(x, y) \leq \delta(x, z) + \delta(z, y).$$

Weak metrics generalize ordinary metrics by allowing asymmetry and zero distances between distinct points. That is, $\delta(x, y)$ and $\delta(y, x)$ may differ, and $\delta(x, y)$ may vanish even when $x \neq y$.

Examples of weak metrics are found in [4]. Some fixed point problems in weak metric spaces were investigated in [3].

The following theorem characterizes convergence in $\sigma(X, X^\diamondsuit)$ in terms of weak metrics.

Theorem 1.6. *Let (X, d) be a metric space and $X^\diamondsuit$ the space of all metric functionals on X. Then, there exists a family of weak metrics $\{\delta_{\mathbf{h}} \mid \mathbf{h} \in X^\diamondsuit\}$ on X such that a net $(x_i)_{i \in I}$ in X converges to $x \in X$ in $\sigma(X, X^\diamondsuit)$ if and only if $\delta_{\mathbf{h}}(x, x_i) \to 0$ for every $\mathbf{h} \in X^\diamondsuit$.*

We conclude the presentation of our main results with some further statements concerning the topology $\sigma(X, X^\diamondsuit)$. The first establishes its separation properties, while the remaining ones compare it with the classical weak topology on normed linear spaces.

Theorem 1.7. *Let X be a metric space. Then $\sigma(X, X^\diamondsuit)$ is T_1. Moreover, if X is a normed linear space, then $\sigma(X, X^\diamondsuit)$ is Hausdorff.*

Theorem 1.8. *Let X be a normed linear space. Then*

$$\sigma(X, X^\diamondsuit) \subseteq \sigma(X, X^*).$$

The inclusion shown above can be either an equality or a strict one, as the following results show.

Corollary 1.9. *Let X be a normed linear space whose dual X^* is strictly convex. Then*

$$\sigma(X, X^\diamondsuit) = \sigma(X, X^*).$$

Proposition 1.10. *Let $X = \ell_1$. Then, we have*

$$\sigma(X, X^\diamondsuit) \subsetneq \sigma(X, X^*).$$

2. Proofs

Let us begin by recalling the notion of a metric functional. Let (X, d) be a metric space. Choose a base point $o \in X$ and define

$$X^\vee := \{\mathbf{h}_w \mid w \in X\}, \quad \text{where} \quad \mathbf{h}_w := d(\cdot, w) - d(o, w).$$

Note that each element in $X^\vee$ is a 1-Lipschitz mapping $X \to \mathbb{R}$ vanishing at the point o. Let $X^\diamondsuit$ denote the closure of $X^\vee$ in the topology of pointwise convergence. Each element in $X^\diamondsuit$ is called a metric functional on X, and each element in $X^\vee$ is called an internal metric functional. For more details see [2].

2.1. **Construction of the weak topology.** We now turn to the construction of the weak topology on X. For a point $x \in X$, a nonempty finite subset $F \subseteq X^\diamondsuit$ and a real number $c > 0$, define

$$B(x, F, c) := \bigcap_{\mathbf{h} \in F} \{y \in X \mid \mathbf{h}(y) > \mathbf{h}(x) - c\}.$$

Lemma 2.1. *The collection*

$$\mathscr{B} := \{B(x, F, c) \mid x \in X,\ F \subseteq X^\diamondsuit,\ 0 < \operatorname{card}(F) < \infty,\ c > 0\}$$

is a basis for a topology on X.

Proof. First, note that every point $x \in X$ is contained in $B(x, \{\mathbf{h}_x\}, 1)$, where

$$\mathbf{h}_x := d(\cdot, x) - d(o, x) \in X^\vee.$$

Now, suppose that x is a point in the intersection of two elements of $\mathscr{B}$, say

$$B(x_1, F_1, c_1) \qquad \text{and} \qquad B(x_2, F_2, c_2).$$

Then, we have

$$\mathbf{h}(x) > \mathbf{h}(x_1) - c_1, \quad \text{for all } \mathbf{h} \in F_1,$$

and

$$\mathbf{h}(x) > \mathbf{h}(x_2) - c_2, \quad \text{for all } \mathbf{h} \in F_2.$$

Define $F := F_1 \cup F_2$ and

$$c := \min\left\{\min_{\mathbf{h} \in F_1}\{\mathbf{h}(x) - \mathbf{h}(x_1) + c_1\},\ \min_{\mathbf{h} \in F_2}\{\mathbf{h}(x) - \mathbf{h}(x_2) + c_2\}\right\}.$$

Since F_1 and F_2 are nonempty finite subsets of $X^\diamondsuit$, so is F. By the preceding inequalities, we have $c > 0$. Hence, $B(x, F, c)$ belongs to $\mathscr{B}$ and satisfies

$$x \in B(x, F, c) \subseteq B(x_1, F_1, c_1) \cap B(x_2, F_2, c_2).$$

□

Definition 2.2. We denote by $\sigma(X, X^\diamondsuit)$ the topology generated by the basis $\mathscr{B}$. That is, a subset $A \subseteq X$ is open in $\sigma(X, X^\diamondsuit)$ if and only if for every $x \in A$ there exists a basis element $B \in \mathscr{B}$ such that $x \in B \subseteq A$.

Note that a subset $A \subseteq X$ is open in $\sigma(X, X^\diamondsuit)$ if and only if for every $x \in A$ there exists a nonempty finite subset $F \subseteq X^\diamondsuit$ and a real number $c > 0$ such that $B(x, F, c) \subseteq A$.

One may wonder if different base points produce different topologies. We show next that the choice of the base point is irrelevant.

Proposition 2.3. *Let p be a point in X such that $d(o,p) > 0$. Let $X_p^\diamondsuit$ denote the space of all metric functionals on X with base point p. Then, we have*

$$\sigma(X, X^\diamondsuit) = \sigma(X, X_p^\diamondsuit).$$

Proof. For every $x, y \in X$ we have

$$d(x,y) - d(o,y) - (d(p,y) - d(o,y)) = d(x,y) - d(p,y).$$

Then, $\mathbf{h} - \mathbf{h}(p)$ is in $X_p^\diamondsuit$ for all $\mathbf{h} \in X^\diamondsuit$. Now, if the points o and p are permuted in the previous equality, we conclude that the mapping $\mathbf{h} \mapsto \mathbf{h} - \mathbf{h}(p)$ is a bijection from $X^\diamondsuit$ onto $X_p^\diamondsuit$.

Since subtracting a constant does not affect the inequalities defining the elements of the basis $\mathscr{B}$, the two topologies coincide. □

Lemma 2.4. *The topology $\sigma(X, X^\diamondsuit)$ is coarser than the metric topology on (X, d).*

Proof. Let A be an open set in the topology $\sigma(X, X^\diamondsuit)$ and $x \in A$. Then, there exists a basis element $B(x, F, c) \subseteq A$. Since every metric functional is 1-Lipschitz, we have

$$U(x,c) := \{y \in X \mid d(x,y) < c\} \subseteq B(x, F, c).$$

Since $U(x,c)$ is an open metric ball, it follows that A is open in the metric topology. □

Metric functionals are continuous as mappings from X to $\mathbb{R}$ with respect to their metric topologies. When X is equipped with the topology $\sigma(X, X^\diamondsuit)$, we have the following weaker property.

Lemma 2.5. *The topology $\sigma(X, X^\diamondsuit)$ makes every metric functional lower semi-continuous on X.*

Proof. Let $\mathbf{h}$ be a metric functional on X. For every real number r, the set

$$A_r := \{y \in X \mid \mathbf{h}(y) > r\}$$

is open in the topology $\sigma(X, X^\diamondsuit)$. Indeed, for every $x \in A_r$, we have

$$B(x, \{\mathbf{h}\}, \mathbf{h}(x) - r) \subseteq A_r.$$

□

We are now ready to complete the proof of Theorem 1.2.

Proof of Theorem 1.2. The first property is shown in Lemma 2.4 and the second property in Lemma 2.5. It remains to prove the third assertion.

Suppose that a net $(a_i)_{i \in I}$, indexed by a directed set $(I, \succcurlyeq)$, converges d-weakly to a point $a \in X$. Let A be an open set in the topology $\sigma(X, X^\diamondsuit)$ with $a \in A$. Then, there exists a real number $c > 0$ and a nonempty finite subset $F \subseteq X^\diamondsuit$ such that

$$B(a, F, c) \subseteq A.$$

By d-weak convergence, for each metric functional $\mathbf{h} \in F$ there exists an index $i_0(c, \mathbf{h}) \in I$ such that

$$\mathbf{h}(a_i) > \mathbf{h}(a) - c,$$

for all $i \succcurlyeq i_0(c, \mathbf{h})$. Since F is a nonempty finite set, there exists an index $i_* \in I$ such that $i_* \succcurlyeq i_0(c, \mathbf{h})$ for all $\mathbf{h} \in F$. Thus, for every $i \succcurlyeq i_*$, we have

$$a_i \in B(a, F, c) \subseteq A.$$

This shows that $(a_i)_{i\in I}$ converges to a in the topology $\sigma(X, X^{\diamondsuit})$.

Now, suppose that a net $(a_i)_{i\in I}$ converges to a point $a \in X$ in the topology $\sigma(X, X^{\diamondsuit})$. Let $\mathbf{h}$ be a metric functional on X and c a positive real number. Since $B(a, \{\mathbf{h}\}, c)$ is open in $\sigma(X, X^{\diamondsuit})$ and contains the point a, there exists an index $i_0 \in I$ such that

$$a_i \in B(a, \{\mathbf{h}\}, c),$$

for all $i \succcurlyeq i_0$. Thus, we have

$$\liminf_{i\in I} \mathbf{h}(a_i) \geq \mathbf{h}(a).$$

□

2.2. **Escaping sequence that converges weakly.** We next establish Theorem 1.3 by constructing an unbounded d-weakly convergent sequence in $C[0, 1]$.

Proof of Theorem 1.3. Choose a sequence of nondegenerate closed intervals

$$A_n \subseteq [0, 1] \quad \text{with} \quad A_n \cap A_m = \varnothing \quad \text{for all } n \neq m.$$

For each $n \geq 1$, choose a nonnegative continuous function f_n on $[0, 1]$ that vanishes outside A_n and attains its maximum value n. Then, for all $n \geq 1$,

$$\|f_n\|_\infty = n.$$

We show next that for every $\mathbf{h} \in C[0, 1]^{\diamondsuit}$ we have

$$\liminf_n \mathbf{h}(f_n) \geq 0.$$

First, we claim that for each internal $\mathbf{h}_w \in C[0, 1]^{\vee}$, where

$$\mathbf{h}_w := \|\cdot - w\|_\infty - \|w\|_\infty,$$

the value $\mathbf{h}_w(f_n)$ is negative for at most one n. Indeed, for a fixed $w \in C[0, 1]$ there exists $s \in [0, 1]$ such that $|w(s)| = \|w\|_\infty$. If $s \notin A_n$, then $f_n(s) = 0$ and

$$\mathbf{h}_w(f_n) = \|f_n - w\|_\infty - \|w\|_\infty \geq |w(s)| - |w(s)| = 0.$$

Since A_n and A_m are disjoint for all $n \neq m$, the value $\mathbf{h}_w(f_n)$ is negative for at most one n.

Now, let $\mathbf{h}$ be an arbitrary metric functional on $C[0, 1]$. Then, there exists a net $(w_i)_{i\in I}$ in $C[0, 1]$ such that $\mathbf{h}_{w_i}(f) \to \mathbf{h}(f)$ for all $f \in C[0, 1]$. Suppose that there exist two distinct positive integers n and m such that

$$\mathbf{h}(f_n) < 0 \quad \text{and} \quad \mathbf{h}(f_m) < 0.$$

Choose $\epsilon > 0$ such that

$$\mathbf{h}(f_n) < -2\epsilon \quad \text{and} \quad \mathbf{h}(f_m) < -2\epsilon.$$

Since $\mathbf{h}_{w_i} \to \mathbf{h}$ pointwise, there exist $j(\epsilon, n)$ and $k(\epsilon, m)$ in the directed set $(I, \succcurlyeq)$ such that

$$\mathbf{h}_{w_i}(f_n) < \mathbf{h}(f_n) + \epsilon \quad \text{for all } i \succcurlyeq j(\epsilon, n)$$

and

$$\mathbf{h}_{w_i}(f_m) < \mathbf{h}(f_m) + \epsilon \quad \text{for all } i \succcurlyeq k(\epsilon, m).$$

Choose $i_0 \in I$ such that $i_0 \succcurlyeq j(\epsilon, n)$ and $i_0 \succcurlyeq k(\epsilon, m)$. Then, for every $i \succcurlyeq i_0$,

$$\max\{\mathbf{h}_{w_i}(f_n), \mathbf{h}_{w_i}(f_m)\} < -\epsilon\,.$$

This contradicts the preceding argument, which shows that each internal metric functional $\mathbf{h}_{w_i}$ can take a negative value for at most one f_n. Therefore, $\mathbf{h}(f_n) < 0$ for at most one n. Consequently,

$$\liminf_n \mathbf{h}(f_n) \geq 0 = \mathbf{h}(0).$$

□

Remark 2.6. We listed in [2, Theorem 1.4] some normed linear spaces in which d-weak convergence of sequences implies boundedness. However, our argument for the case $C[0,1]$ contains an error, as demonstrated by the sequence (f_n) constructed above.

In ℓ_1 and in normed linear spaces with strictly convex duals, d-weakly convergent sequences are bounded.

2.3. **Invariance under surjective isometries.** We now turn to the invariance of the topology $\sigma(X, X^\diamondsuit)$ under surjective isometries.

Proof of Theorem 1.4. Choose $o \in X$ and $To \in Y$ as the base points in the constructions of $X^\diamondsuit$ and $Y^\diamondsuit$, respectively. Since T is surjective and distance-preserving, for every point $y \in Y$ there exists a point $w \in X$ such that $y = Tw$ and

$$D(Tx, y) - D(To, y) = d(x, w) - d(o, w) \quad \text{for all } x \in X.$$

Then, the mapping

$$T^\diamondsuit : Y^\diamondsuit \to X^\diamondsuit \quad \mathbf{h} \mapsto \mathbf{h} \circ T$$

is a bijection whose inverse is given by

$$(T^\diamondsuit)^{-1}(\mathbf{h}) = \mathbf{h} \circ T^{-1} \quad \text{for all } \mathbf{h} \in X^\diamondsuit.$$

If a net $(x_i)_{i\in I}$ in X converges d-weakly to $x \in X$, then for every $\mathbf{h} \in Y^\diamondsuit$,

$$\begin{aligned} \liminf_{i\in I} \mathbf{h}(Tx_i) &= \liminf_{i\in I} T^\diamondsuit \mathbf{h}(x_i) \\ &\geq T^\diamondsuit \mathbf{h}(x) \\ &= \mathbf{h}(Tx). \end{aligned}$$

Thus $(Tx_i)_{i\in I}$ converges D-weakly to $Tx \in Y$.

Conversely, if a net $(y_i)_{i\in I}$ in Y converges D-weakly to $y \in Y$, then for every $\mathbf{h} \in X^\diamondsuit$,

$$\begin{aligned} \liminf_{i\in I} \mathbf{h}(T^{-1}y_i) &= \liminf_{i\in I} ((T^\diamondsuit)^{-1} \mathbf{h})(y_i) \\ &\geq ((T^\diamondsuit)^{-1} \mathbf{h})(y) \\ &= \mathbf{h}(T^{-1}y). \end{aligned}$$

Hence $(T^{-1}y_i)_{i\in I}$ converges d-weakly to $T^{-1}y \in X$.

By Theorem 1.2, the preceding arguments show that T is a homeomorphism between $(X, \sigma(X, X^\diamondsuit))$ and $(Y, \sigma(Y, Y^\diamondsuit))$. □

2.4. **Weak metrics.** For each $\mathbf{h} \in X^\diamondsuit$ define the mapping $\delta_{\mathbf{h}} : X \times X \to [0, +\infty)$ by

$$(x, y) \mapsto \delta_{\mathbf{h}}(x, y) \coloneqq \max\{0, \mathbf{h}(x) - \mathbf{h}(y)\}.$$

Lemma 2.7. *For every $\mathbf{h} \in X^\diamondsuit$, $\delta_{\mathbf{h}}$ is a weak metric on X.*

Proof. Recall that for every pair of real numbers a and b we have

$$0 \leq \max\{0, a + b\} \leq \max\{0, a\} + \max\{0, b\}.$$

□

Proof of Theorem 1.6. By Theorem 1.2, a net $(x_i)_{i \in I}$ in X converges to $x \in X$ in the topology $\sigma(X, X^\diamondsuit)$ if and only if for every $\mathbf{h} \in X^\diamondsuit$ and every real number $\epsilon > 0$ there exists an index $i_0 \in I$ such that

$$\mathbf{h}(x_i) > \mathbf{h}(x) - \epsilon \qquad \text{for all } i \succcurlyeq i_0.$$

This is equivalent to the property $\delta_{\mathbf{h}}(x, x_i) \to 0$ for all $\mathbf{h} \in X^\diamondsuit$. □

2.5. **Separation properties.**

Proof of Theorem 1.7. Let (X, d) be a metric space and let $x, y \in X$ be distinct points. Recall that

$$\mathbf{h}_x, \mathbf{h}_y \in X^\vee \subseteq X^\diamondsuit.$$

Since $\mathbf{h}_y(x) - \mathbf{h}_y(y) = d(x, y) > 0$, we have

$$y \notin B(x, \{\mathbf{h}_y\}, d(x, y)).$$

Similarly, $\mathbf{h}_x(y) - \mathbf{h}_x(x) = d(x, y) > 0$ implies

$$x \notin B(y, \{\mathbf{h}_x\}, d(x, y)).$$

Thus the topology $\sigma(X, X^\diamondsuit)$ is T_1.

Suppose now that X is a normed linear space. By the Hahn–Banach and Krein–Milman theorems, the extreme points of the closed unit ball of X^* separate points of X. Thus, for distinct $x, y \in X$, there exists an extreme point f such that $f(x) < f(y)$. Since the dual unit ball is symmetric, $-f$ is also an extreme point. By [6, Corollary 3.5], both f and $-f$ belong to $X^\diamondsuit$.

Define

$$c \coloneqq \frac{f(y) - f(x)}{2} > 0.$$

Then

$$B(x, \{-f\}, c) = \{z \in X \mid f(z) < f(x) + c\}$$

and

$$B(y, \{f\}, c) = \{z \in X \mid f(z) > f(y) - c\}$$

are open neighborhoods of x and y, respectively. Since

$$f(x) + c = f(y) - c,$$

these neighborhoods are disjoint. Hence $\sigma(X, X^\diamondsuit)$ is Hausdorff. □

The topology $\sigma(X, X^\diamondsuit)$ need not be Hausdorff for general metric spaces. For example, let $X = \mathbb{R}$ be equipped with the metric

$$d(x, y) \coloneqq \sqrt{|x - y|}.$$

Then $X^\diamondsuit = X^\vee \cup \{\mathbf{0}\}$, and the sequence (a_n) defined by $a_n \coloneqq n$ converges to every point of X in the topology $\sigma(X, X^\diamondsuit)$.

2.6. **Comparison with the classical weak topology.**

Proof of Theorem 1.8. Every metric functional is convex, because internal metric functionals are convex and convexity is preserved under pointwise limits. Moreover, every metric functional is 1-Lipschitz and therefore norm-continuous. Consequently, for each $\mathbf{h} \in X^{\diamondsuit}$ and $a \in \mathbb{R}$, the set

$$\{x \in X \mid \mathbf{h}(x) \leq a\}$$

is convex and norm-closed, hence closed in the classical weak topology $\sigma(X, X^*)$ by the Hahn–Banach separation theorem. Thus, the set $\{x \in X \mid \mathbf{h}(x) > a\}$ is open in $\sigma(X, X^*)$. Since strict superlevel sets of metric functionals generate the topology $\sigma(X, X^{\diamondsuit})$, the asserted inclusion follows. □

Proof of Corollary 1.9. By Theorem 1.8, it suffices to prove the inclusion

$$\sigma(X, X^*) \subseteq \sigma(X, X^{\diamondsuit}).$$

Since X^* is strictly convex, every element of its unit sphere is an extreme point of its closed unit ball. By [6, Corollary 3.5], every such functional belongs to $X^{\diamondsuit}$. Consequently, every element of X^* is continuous with respect to $\sigma(X, X^{\diamondsuit})$. Thus

$$\sigma(X, X^*) \subseteq \sigma(X, X^{\diamondsuit}).$$

□

Proof of Proposition 1.10. By [1, Theorem 3.6], every $\mathbf{h} \in X^{\diamondsuit}$ has the form

$$\mathbf{h}(x) = \sum_{n \in I} \varepsilon_n x_n + \sum_{n \notin I} \big(|x_n - z_n| - |z_n|\big),$$

where $I \subseteq \mathbb{N}$, $\varepsilon_n \in \{-1, 1\}$ for $n \in I$, and $z_n \in \mathbb{R}$ for $n \notin I$.

For each $\mathbf{h} \in X^{\diamondsuit}$, define $s = (s_n) \in \{-1, 1\}^{\mathbb{N}}$ by

$$s_n := \begin{cases} \varepsilon_n, & n \in I, \\ -\operatorname{sgn}(z_n), & n \notin I,\ z_n \neq 0, \\ 1, & n \notin I,\ z_n = 0. \end{cases}$$

Since, for all $a, b \in \mathbb{R}$ with $b \neq 0$,

$$|a - b| \geq \operatorname{sgn}(b)(b - a) = |b| - \operatorname{sgn}(b)a,$$

and $|a| \geq a$, it follows that, for every $x \in X$,

$$\mathbf{h}(x) \geq \sum_{n=1}^{\infty} s_n x_n.$$

We now show that the functional $f \in X^*$ given by

$$f(x) = \sum_{n=1}^{\infty} \frac{x_n}{n}$$

is not continuous in the topology $\sigma(X, X^{\diamondsuit})$. Suppose otherwise. Define

$$U := \{x \in X \mid f(x) < 1\}.$$

Then U is an open neighborhood of 0 in the topology $\sigma(X, X^{\diamondsuit})$. Hence there exist

$$\mathbf{h}^1, \ldots, \mathbf{h}^m \in X^{\diamondsuit} \quad \text{and} \quad c > 0$$

such that

$$B(0, \{\mathbf{h}^1, \dots, \mathbf{h}^m\}, c) \subseteq U.$$

For each $1 \leq j \leq m$, choose a sign sequence $s^j = (s_n^j)$ satisfying

$$\mathbf{h}^j(x) \geq \sum_{n=1}^{\infty} s_n^j x_n.$$

for all $x \in X$. Since the sequence

$$v_n \coloneqq (s_n^1, \dots, s_n^m), \qquad n \in \mathbb{N},$$

takes values in the finite set $\{-1, 1\}^m$, there exist distinct $p, q \in \mathbb{N}$ such that $v_p = v_q$, that is,

$$s_p^j = s_q^j \qquad \text{for } j = 1, \dots, m.$$

Let (e_n) be the canonical basis of X and define

$$x \coloneqq \frac{e_p - e_q}{1/p - 1/q}.$$

Then $f(x) = 1$, so $x \notin U$. On the other hand, for every $j = 1, \dots, m$, we have

$$\mathbf{h}^j(x) \geq \frac{s_p^j - s_q^j}{1/p - 1/q} = 0 > -c.$$

Thus

$$x \in B(0, \{\mathbf{h}^1, \dots, \mathbf{h}^m\}, c) \subseteq U,$$

a contradiction.

Therefore f is not continuous in the topology $\sigma(X, X^\diamondsuit)$, hence we have

$$\sigma(X, X^\diamondsuit) \subsetneq \sigma(X, X^*).$$

□

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